DevSecops Linux Assignment

Scenario 1 – Part A

```
Ubuntu                    Ubuntu
root@Khronicles-PC:~# useradd -m -s /bin/bash audit_test
root@Khronicles-PC:~# ls -al /home
total 16
drwxr-xr-x  4 root       root       4096 Feb 20 18:16 .
drwxr-xr-x 22 root       root       4096 Feb 20 14:33 ..
drwxr-x---  2 audit_test audit_test 4096 Feb 20 18:16 audit_test
drwxr-x---  4 khroncles  khroncles  4096 Feb 20 18:12 khroncles
root@Khronicles-PC:~#
```

Part – B

The second column usually shows an encrypted string which indicates the user has a password set. However, from the image, audit_test has an exclamation mark indicating this user does not have a password set.

```
root@Khronicles-PC:~# less /etc/shadow
root@Khronicles-PC:~#
```

```
khroncles:$y$j9T$f8e3zu61Ru3TBikBP3MNW0$QfCOFTT6WCRr2gjudcYBObQk0/4/bmB.wFxw1f9UMG1:20504:0:
99999:7:::
audit_test:!:20504:0:99999:7:::
/etc/shadow (END)
```

Part – C

Password setting command:

```
root@Khronicles-PC:~# passwd audit_test
New password:
Retype new password:
passwd: password updated successfully
root@Khronicles-PC:~#
```

Updated password before and after:

```
khroncles:$y$j9T$f8e3zu61Ru3TBikBP3MNW0$QfCOFTT6WCRr2gjudcYBObQk0/4/bmB.wFxw1f9UMG1:20504:0:
99999:7:::
audit_test:!:20504:0:99999:7:::
/etc/shadow (END)
```

```
audit_test:$y$j9T$RXLYPomkHpmilK/PiDwUS.$MSIrOymeKs83NYHVrm0PITOZTU8ypwNSU4kOAAkolr4:20504:0
:99999:7:::
/etc/shadow (END)
```

Scenario 2

User Onboarding:

```
Ubuntu    Ubuntu
root@Khronicles-PC:~# useradd -m -s /bin/bash lateef && passwd lateef
New password:
Retype new password:
passwd: password updated successfully
root@Khronicles-PC:~# useradd -m -s /bin/bash purity && passwd purity
New password:
Retype new password:
passwd: password updated successfully
root@Khronicles-PC:~# useradd -m -s /bin/bash arthur && passwd arthur
New password:
Retype new password:
passwd: password updated successfully
root@Khronicles-PC:~# useradd -m -s /bin/bash paula && passwd paula
New password:
Retype new password:
passwd: password updated successfully
root@Khronicles-PC:~# useradd -m -s /bin/bash yaa && passwd yaa
New password:
Retype new password:
passwd: password updated successfully
root@Khronicles-PC:~# useradd -m -s /bin/bash alex && passwd alex
New password:
Retype new password:
passwd: password updated successfully
root@Khronicles-PC:~# useradd -m -s /bin/bash habiba
root@Khronicles-PC:~# useradd -m -s /bin/bash aminat && passwd aminat
New password:
Retype new password:
passwd: password updated successfully
root@Khronicles-PC:~#
```

Groups Onboarding:

```
root@Khronicles-PC:~# groupadd dev_team && groupadd sec_team && groupadd qa_team && groupadd
ops_team
groupaddops_team: command not found
root@Khronicles-PC:~# groupadd ops_team
root@Khronicles-PC:~#
```

Non-login account Onboarding with configured non-login shell:

```
root@Khronicles-PC:~# adduser ci_runner --system --shell /usr/sbin/nologin
info: Selecting UID from range 100 to 999 ...

info: Adding system user `ci_runner' (UID 105) ...
info: Adding new user `ci_runner' (UID 105) with group `nogroup' ...
info: Not creating `/nonexistent'.
```

User-Group Assignment:

```
root@Khronicles-PC:~# usermod -aG dev_team lateef && usermod -aG dev_team arthur
root@Khronicles-PC:~# usermod -aG sec_team habiba && usermod -aG sec_team purity
root@Khronicles-PC:~# usermod -aG qa_team paula && usermod -aG qa_team yaa
root@Khronicles-PC:~# usermod -aG ops_team aminat && usermod -aG ops_team alex
```

Habiba’s account before password:

```
habiba:!:20504:0:99999:7:::
aminat:$y$j9T$vUjcJR62uR5huKytwkJ0k.$jrJ5hA/LIwF
99:7:::
ci_runner:!:20504:::::
/etc/shadow (END)
```

Habiba’s account after password:

```
root@Khronicles-PC:~# passwd habiba
New password:
Retype new password:
passwd: password updated successfully
root@Khronicles-PC:~#
```

```
habiba:$y$j9T$hVrOKu0RCk6Xq012rmQWb.$HpTRqTTb9mc.hBkwQ3RXeSMDel9fKx5lonVaHnKLXr3:20504:0:999
99:7:::
```

Sudo Access:

```
root@Khronicles-PC:~# ls -al /etc/sudoers.d
total 16
drwxr-xr-x  2 root root 4096 Feb 20 21:55 .
drwxr-xr-x 88 root root 4096 Feb 20 21:56 ..
-r--r-----  1 root root 1068 Jan 29  2024 README
-rw-r-----  1 root root   20 Feb 20 21:55 ops_team
```

Sudo Access Verification (Access granted to Only Alex):

```
Ubuntu                    Ubuntu

alex ALL=(ALL) ALL
/etc/sudoers.d/ops_team (END)
```

```
Matching Defaults entries for alex on Khronicles-PC:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin
,
    use_pty

User alex may run the following commands on Khronicles-PC:
    (ALL) ALL
root@Khronicles-PC:~#
```

Removed user (Aminat with preserved home dir):

```
root@Khronicles-PC:~# userdel aminat
root@Khronicles-PC:~# ls /home
alex  aminat  arthur  audit_test  habiba  khroncles  lateef  paula  purity  yaa
root@Khronicles-PC:~#
```

Group data evidence (file is /etc/group):

```
dev_team:x:1010:lateef,arthur
sec_team:x:1011:habiba,purity
qa_team:x:1012:paula,yaa
ops_team:x:1013:alex
(END)
```

Explanations:

1. Linux records group membership in system files (/etc/passwd, /etc/group, /etc/shadow), tracks it through UID/GID mappings, and enforces it at runtime through commands.
2. Linux enforces department-based access control by assigning users to department groups, linking resources to those groups and using file permissions and sudo rules to ensure only authorized departments can access sensitive data.
3. Service accounts must use non-login shells to prevent interactive logins and reduce security risks.
4. Home directories may be retained after user deletion to preserve important files and data the user created.
5. DevSecOps controls privilege escalation by combining least privilege, RBAC, policy enforcement, auditing, and time-bound access, ensuring that elevated rights are only granted when justified and traceable.